Course Code : ECT 355-4 **KQLR/RS – 24 / 1610**

Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

SMART SENSORS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.
(2) Illustrate your answers with a neat diagram wherever necessary.
(3) All questions are compulsory.

1. (a) What is effect of environmental factors on performance of sensors ? What do you understand by pre-ageing of a sensor ? 6(CO1)

   (b) How does Resolution contrast with Accuracy ? Illustrate with an example. 4(CO1)

2. (a) Due to certain working conditions assume that a tractor engine is heated-up more than 350^{0}C. For measurement of its temperature only two choices are available, either RTD or Thermistor sensors. Based on your understanding. How would you determine the sensor to be used ? 5(CO2)

   (b) What are different micromachining processes used in fabrication of micromechanical components ? 5(CO4)

3. (a) Analyze the fabrication technique that is useful in producing the sensor parts of heights ranging from micrometers to centimeters. 5(CO4)

   (b) What is purpose of TEDS in IEEE 1451 standards ? Discuss its benefits. 5(CO3)

4. (a) What is purpose of IEEE 1451 standards for Smart Sensors ? Can you write a brief outline on IEEE 1451 standards ? 5(CO3)

   (b) Explain in brief, ceramic as a sensor material. 5(CO2)

5. (a) Plastics are typically insulators. Justify the use of plastic as a sensor material. 5(CO2)

   (b) Discuss Chemical Vapor Deposition (CVD) process and its use. 5(CO5)

6. (a) How can you justify use of Smart Sensors in Smart cars ? What compels industry to use smart sensors in this application in place of traditional sensors ? 5(CO5)

   (b) What properties a Smart Sensor must possess if it is to be used in automated consumer applications such as smart cars or smart homes ? 5(CO5)